

Renewable Intermittency and Grid Reliability

An Endogeneous Learning Model for Vietnam

Midterm Progress Report

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Spring 2026

Presentation Outline

- 1 Motivation & Research Questions
- 2 Model Environment
- 3 Model Mechanics & Log-Linearization
- 4 Preliminary Results
- 5 Work in Progress

Vietnam's Energy Trilemma

Vietnam's growth model confronts three simultaneously binding constraints:

- **Energy security:** rapidly rising industrial demand
- **Sustainability:** Paris Agreement + Net Zero pledges
- **Affordability:** administratively regulated retail tariffs (EVN)

PDP8 at a Glance

	2024 actual	2030 target
Solar + Wind	22.8 GW	73.5 GW
Battery BESS	0.25 GW	14.0 GW
Grid invest.	\$3.2 B/yr	\$3.6 B/yr

The Core Structural Friction

Renewable generation is *non-dispatchable*: supply clears in real time with demand. As VRE penetration rises without adequate storage, the **reliability penalty**—endogenous reduction in capital utilization from grid instability—acts like a persistent negative TFP shock.

Empirical Trigger

Solar capacity doubled in 18 months (2020–21) under FiT incentives, while battery storage remained near zero. Vietnam already experiences curtailment >10% in southern provinces.

Literature Gap: What Existing Models Miss

What the literature covers:

- *E-DSGE models* (Heutel 2012; Airaud et al. 2025): carbon taxes, “greenflation”
- *Directed Technical Change* (Acemoglu et al. 2012): R&D-race between sectors
- *Techno-economic* (Zakeri & Syri 2015; Schreiner 2020): storage costs, grid investment
- *Open economy* (Schmitt-Grohé 2003): debt-elastic interest rates

Critical blind spots:

- No model *endogenizes grid reliability* as a capital utilization constraint
- DTC literature ignores *deployment-driven learning* in storage
- Public–private investment *asymmetry* (“agility gap”) is not formalized
- Open-economy E-DSGE ignores *battery import price* as a terms-of-trade shock

This capstone bridges all four gaps jointly—in a single DSGE calibrated to Vietnam.

Research Questions & Contributions

Four Research Questions

- ① How does VRE intermittency propagate through capital utilization into macroeconomic output losses?
- ② What is the structural timing mismatch between private battery and public grid investment—the **Agility Gap**?
- ③ How does endogenous learning-by-doing in storage depend on regulatory price signal transmission (χ)?
- ④ What are the welfare costs of intermittency and how do policy instruments rank in closing the gap?

Contribution 1

Endogenous reliability function:
 $u_t = f(\text{flexibility/VRE vol})$

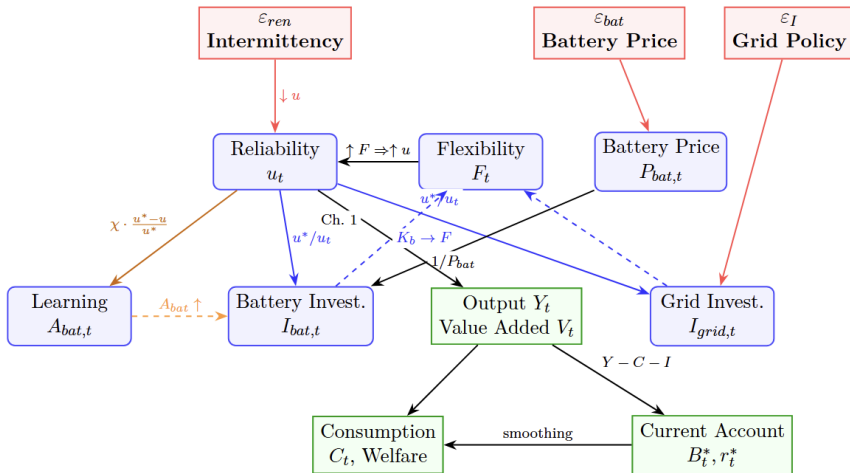
Contribution 2

Reliability-gap learning: ΔA_{bat}
 responds to scarcity signal χ

Contribution 3

CES agility gap: private batteries (agile) vs. public grid (sluggish)

Model Architecture: Small Open Economy DSGE



Households

Objective

- Representative household maximizes discounted utility over consumption C_t and labor L_t
- Standard log utility in consumption; disutility of labor
- Discount factor $\beta = 0.990$ (4% annualized real return)

Budget Constraint

- Income: wages $w_t L_t$, capital rental $R_t K_{p,t}$, foreign bond returns
- Expenditure: consumption, productive investment, **battery purchases** $P_{bat,t} I_{bat,t}$, taxes, foreign bonds

Optimality Conditions

- **Capital Euler:** equates marginal cost of saving today to discounted return on capital tomorrow
- **Bond Euler:** links domestic consumption growth to world interest rate r_t^*
- **Labor supply:** wages equal marginal disutility of work

Key modeling choice: Capital utilization u_t is *not* a household decision. It is an exogenous physical constraint set by grid reliability—not a margin households can optimize over.

Production & The Two Novel Equations

- Output is a CES aggregate of value-added V_t and energy $\bar{E}$ (substitutability $\sigma_E = 0.6$)
- Value-added is Cobb-Douglas with capital share $\alpha = 0.35$ — but effective capital is $u_t K_{p,t}$, so grid reliability directly reduces productive capacity

Contribution 1 — Reliability Constraint

$$u_t = 1 - \exp\left(-\psi \frac{F_t}{\phi_{int} \cdot \text{Vol}_{ren,t}}\right)$$

Utilization rises as the *flexibility stock* F_t grows relative to VRE volatility. The exponential form captures nonlinear reliability costs. When $F_t \uparrow$ or volatility $\downarrow$, $u_t \rightarrow 1$.

Contribution 2 — CES Flexibility Bundle

$$F_t = \left[\mu (A_{bat,t} K_{b,t})^{\frac{\rho_f - 1}{\rho_f}} + (1 - \mu) K_{g,t}^{\frac{\rho_f - 1}{\rho_f}} \right]^{\frac{\rho_f}{\rho_f - 1}}$$

Batteries and grid are *complements* ($\rho_f = 0.4 < 1$): neither can substitute for the other when the opposing asset is scarce. $A_{bat,t}$ captures endogenous learning.

Battery Investment & Endogenous Learning

Battery Investment (Private, Agile)

- Investment responds *contemporaneously* to the reliability gap ($u^* - u_t$): the larger the shortfall, the more private capital is deployed
- Dampened by the world battery price $P_{bat,t}$ — Vietnam is a price-taker on global lithium markets
- Capital accumulates each period net of 12% annual depreciation (Li-ion lifecycle)

Contribution 3 — Endogenous Learning

- Battery productivity $A_{bat,t}$ improves when the reliability gap is large: scarcity drives learning-by-doing
- Learning speed $\eta_{bat} = 0.10$ anchored to the 18–22% Li-ion experience curve
- $\chi \in [0, 1]$: regulatory transmission parameter — how well market prices signal scarcity
- Vietnam ($\chi = 0.3$)**: regulated EVN tariffs suppress the scarcity signal, slowing battery cost reduction

Agility Gap: Private batteries respond to reliability shortfalls *immediately*; public grid investment lags 5–7 years due to permitting and construction. This asymmetry generates the mid-transition reliability valley.

Three Transmission Channels

Channel 1 — The Reliability Wedge (Output Loss)

An intermittency shock raises VRE volatility $\Rightarrow$ reliability u_t falls $\Rightarrow$ effective capital $u_t K_{p,t}$ shrinks $\Rightarrow$ output Y_t falls. Acts like a persistent, supply-side negative TFP shock, amplified by energy–capital complementarity.

Channel 2 — The Agility Gap (Asymmetric Investment)

When $u_t < u^*$: private batteries invest *immediately* (agile); public grid investment responds but faces 5–7 year construction lags (sluggish). The timing mismatch generates a predictable mid-transition reliability valley.

Channel 3 — Reliability-Gap Learning (Innovation)

A reliability shortfall triggers endogenous battery cost reduction via learning-by-doing. The speed of learning depends on χ : regulated tariffs suppress scarcity signals, slowing the innovation response and deepening the “valley of death” for storage technology.

Work in Progress

Completed ✓

- Full model derivation (16-equation equilibrium)
- Analytical steady-state characterization & solver
- Dynare implementation & Blanchard-Kahn verification
- Impulse response functions (3 shocks)
- Deterministic transition simulation (Reliability Valley)
- Log-linearization & transmission channel analysis

In Progress / Remaining

- Full calibration (structural + reliability + LBD blocks)
- **Welfare analysis:** consumption-equivalent variation $\Lambda(\chi)$
- **Counterfactual experiments:** battery subsidy ($P_{bat} = 0.8$); accelerated grid ($\phi_{grid} = 3.0$); signal suppression ($\chi = 0.3$)
- **“Perfect Storm” scenario:** joint $\varepsilon_{ren} + \varepsilon_{bat}$ simulation
- **Sensitivity analysis:** $\psi, \mu, \beta, \phi_{grid}$
- **Formal propositions:** prove $\Lambda(\chi)$ monotonicity and CES supermodularity
- **Policy section:** map $\chi \rightarrow$ concrete EVN reform agenda
- **LaTeX manuscript:** finalize bibliography & appendix proofs

Summary: Three Contributions to the Literature

1 Endogenous Grid Reliability as a Macro Friction

Capital utilization $u_t = 1 - \exp(-\psi F_t / \phi_{int} \text{Vol}_{ren,t})$ is determined endogenously by the stock of flexibility assets relative to VRE volatility. This transforms an engineering concept into a macroeconomic wedge in the aggregate production function—a channel absent from all extant E-DSGE models.

2 Reliability-Gap-Induced Learning-by-Doing

Battery technology evolves endogenously: $\Delta A_{bat} = \eta_{bat} \cdot \chi \cdot (u^* - u) / u^*$. Unlike DTC models that require explicit R&D sectors, innovation here responds to *deployment-driven scarcity*. The regulatory parameter χ maps directly to observable policy instruments. **Policy Rec:** Vietnam must establish dynamic capacity markets (raising χ) to unlock storage cost reductions.

3 Identification of the Structural Agility Gap

Private batteries (agile, price-driven) and public grid capital (sluggish, rule-based) are formally distinguished as *complementary* inputs in a CES flexibility bundle ($\rho_f = 0.4$). This generates a predictable mid-transition reliability valley—a quantitative prediction unavailable in models that treat flexibility as homogeneous.

Key References

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Thank You

Renewable Intermittency and Grid Reliability:

An Endogeneous Learning Model of Vietnam

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Keywords: Environmental DSGE · Endogenous Learning · Investment Inertia
Renewable Intermittency · Energy Storage · Vietnam PDP8

Code repository: `vietnam_dsge.mod` + `run_dynare.m` (MATLAB/Dynare 6)

Appendix: Household Equations

Lifetime Utility & Budget Constraint

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \frac{L_t^{1+\sigma_L}}{1+\sigma_L} \right]$$

$$\begin{aligned} C_t + I_{p,t} + P_{bat,t} I_{bat,t} + B_t^* + T_t \\ = w_t L_t + R_t K_{p,t} + (1 + r_{t-1}^*) B_{t-1}^* \end{aligned}$$

Optimality Conditions

Capital Euler:

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left[\frac{\alpha Y_{t+1} / K_{p,t+1} + 1 - \delta_p}{C_{t+1}} \right]$$

Labor Supply:

$$(1 - \alpha) V_t = \varphi C_t L_t^{1+\sigma_L}$$

Appendix: Production Technology

Final output (CES, $\sigma_E = 0.6$):

$$Y_t = [s_V V_t^{(\sigma_E-1)/\sigma_E} + (1 - s_V) \bar{E}^{(\sigma_E-1)/\sigma_E}]^{\sigma_E/(\sigma_E-1)}$$

Value-added (Cobb-Douglas with endogenous utilization):

$$V_t = (u_t K_{p,t})^\alpha L_t^{1-\alpha}, \quad \alpha = 0.35$$

Investment rules:

$$\frac{l_{bat,t}}{\bar{l}_{bat}} = \left(\frac{u^*}{u_t} \right)^{\phi_{grid}} \frac{1}{P_{bat,t}} \quad \hat{l}_{grid,t} = -\phi_{grid} \hat{u}_t + \varepsilon_{l,t}$$

Learning-by-doing:

$$A_{bat,t} - A_{bat,t-1} = \eta_{bat} \cdot \chi \cdot \frac{u^* - u_t}{u^*}$$

Appendix: External Sector Equations

Current account / resource constraint:

$$B_t^* = (1 + r_{t-1}^*)B_{t-1}^* + Y_t - C_t - I_{p,t} - P_{bat,t}I_{bat,t} - I_{grid,t}$$

Debt-elastic interest premium (Schmitt-Grohé 2003):

$$r_t^* = \bar{r} + \phi_b \left(e^{\bar{B}^* - B_t^*} - 1 \right), \quad \phi_b = 0.001$$

Blanchard-Kahn eigenvalues:

$\lambda_1 = 1.033$, $\lambda_2 = -2.229$, $\lambda_3 = \infty$ (3 unstable) matching C_t , Y_t , r_t^*
7 stable eigenvalues: 0, ≈ 0 , 0.90, 0.97, 0.98, 0.99, 0.99 $\Rightarrow$ unique saddle-path $\checkmark$

Appendix: Full System of Log-Linearized Equations

The complete 16-equation linearized system (hat notation: $\hat{x}_t \equiv \ln X_t - \ln X_{ss}$):

① Capital Euler: $\hat{C}_t = \mathbb{E}_t \hat{C}_{t+1} - \frac{\alpha r_{p,ss}}{1+r_{p,ss}} (\mathbb{E}_t \hat{Y}_{t+1} - \mathbb{E}_t \hat{K}_{p,t+1})$

② Bond Euler: $\hat{C}_t = \mathbb{E}_t \hat{C}_{t+1} - \frac{r_{ss}^*}{1+r_{ss}^*} \mathbb{E}_t \hat{r}_{t+1}^*$

③ Labor: $\hat{V}_t = \hat{C}_t + (1 + \sigma_L) \hat{L}_t$

④ CES output: $\hat{Y}_t = s_V^{\sigma} \hat{V}_t$

⑤ Value-added: $\hat{V}_t = \alpha(\hat{u}_t + \hat{K}_{p,t}) + (1 - \alpha) \hat{L}_t$

⑥ Reliability: $\hat{u}_t = -\xi \hat{F}_t + \xi \hat{V}_{ol_{ren},t}$

⑦ Flexibility: $\hat{F}_t = s_b(\hat{A}_{bat,t} + \hat{K}_{b,t}) + (1 - s_b) \hat{K}_{g,t}$

⑧ Battery invest: $\hat{I}_{bat,t} = -\phi_{grid} \hat{u}_t - \hat{P}_{bat,t}$

⑨ Battery capital: $\hat{K}_{b,t+1} = (1 - \delta_b) \hat{K}_{b,t} + \frac{I_{bat,ss}}{K_{b,ss}} \hat{I}_{bat,t}$

⑩ Learning: $\hat{A}_{bat,t} - \hat{A}_{bat,t-1} = -\eta_{bat} \cdot \chi \cdot \hat{u}_t$

⑪ Grid invest: $\hat{I}_{grid,t} = -\phi_{grid} \hat{u}_t + \varepsilon_{I,t}$

⑫ Grid capital: $\hat{K}_{g,t+1} = (1 - \delta_g) \hat{K}_{g,t} + \frac{I_{grid,ss}}{K_{g,ss}} \hat{I}_{grid,t}$

⑬ Prod. capital: $\hat{K}_{p,t+1} = (1 - \delta_p) \hat{K}_{p,t} + \frac{I_{p,ss}}{K_{p,ss}} \hat{I}_{p,t}$

⑭ Current account: [linearized B_t^* equation]

⑮ Interest premium: $\hat{r}_t^* = -\frac{\phi_b}{r_{ss}^*} B_t^*$

⑯ Battery price AR(1): $\hat{P}_{bat,t} = \rho_{bat} \hat{P}_{bat,t-1} + \varepsilon_t^{bat}$

Appendix: Welfare Cost Formula

Consumption-Equivalent Variation Λ (Lucas 1987)

Permanent % reduction in SS consumption leaving the household indifferent between the stochastic and deterministic economies:

$$\Lambda = 1 - \exp \left[\frac{1 - \beta}{1 - \beta^{T+1}} \sum_{t=0}^T \beta^t \log \left(\frac{C_t^{shock}}{C_{ss}} \right) \right]$$

$T = 40$ quarters; C_t^{shock} follows the IRF path.

Proposition: $\Lambda(\chi)$ Strictly Decreasing

$\chi \uparrow \Rightarrow$ faster $A_{bat} \Rightarrow$ higher $F_t, u_t, Y_t, C_t \Rightarrow$ lower Λ .

Proof: monotone comparative statics on linearized innovation equation.

Preliminary Result

χ	Λ	Δ from baseline
1.0 (baseline)	0.000054%	—
0.5 (partial)	0.000063%	+17%
0.0 (suppressed)	0.000073%	+35%

Interpretation Caveat

Λ magnitudes are small because linearization smooths the nonlinear reliability valley. *Relative* rankings across χ are robust. Full nonlinear welfare costs will be computed at the final defense.

Public Grid Investment & External Sector

Grid Investment (Government, Sluggish)

- *Reactive* rule: government invests more when reliability falls below target, with response strength $\phi_{grid} = 1.5$
- Financed by lump-sum taxes; Ricardian equivalence holds — no crowding-out effect
- Implementation delay shock $\varepsilon_{I,t}$ captures budget cycles, permitting, and construction lags (5–7 years)
- Grid capital depreciates at 5% annually (transmission infrastructure)

Small Open Economy (Schmitt-Grohé 2003)

- Vietnam borrows and lends at a world interest rate $\bar{r}$ subject to a debt-elastic premium
- Battery storage is *imported* at world price $P_{bat,t}$ — a direct terms-of-trade shock channel
- **“Twin drain”**: an intermittency shock simultaneously reduces output *and* raises battery import demand, hitting the current account twice
- Higher foreign debt raises the sovereign borrowing rate, feeding back into domestic investment

Exogenous Shocks & Equilibrium

Three Exogenous Driving Forces

- **Intermittency shock** ε_{ren} : AR(1) with persistence $\rho_{ren} = 0.85$ (monsoon seasonality); volatility $\sigma_{ren} = 0.12$ (CV of monthly VN solar output)
- **Battery price shock** ε_{bat} : AR(1) with persistence $\rho_{bat} = 0.90$; volatility $\sigma_{bat} = 0.08$ (lithium carbonate cycle, BNEF 2023)
- **Grid policy shock** ε_I : implementation delays from budget cycles and permitting; mutually independent of other shocks

Competitive Equilibrium: 16 Equations, 16 Unknowns

Block	Equations
Households	Capital Euler, Bond Euler, Labor supply
Production	CES output, Value-added
Energy/Grid	Reliability u_t , Flexibility F_t
Battery	I_{bat} rule, K_b accumulation, LBD A_{bat}
Grid	I_{grid} rule, K_g accumulation
Capital	K_p accumulation
External	Current account, Interest premium
Shocks	Battery price AR(1)

Blanchard-Kahn verified: 3 eigenvalues outside the unit circle match the 3 forward-looking variables (C_t , Y_t , r_t^*). Unique saddle-path equilibrium confirmed.

Calibration Strategy: Three-Stage Approach

- ① **Standard structural parameters** matched to long-run Vietnamese macroeconomic aggregates (GSO 2024, ADB, World Bank)
- ② **Reliability & intermittency parameters** micro-founded on engineering data: NLDC facility-level generation, IEA Vietnam Energy Outlook 2023, EVN annual reliability reports
- ③ **Learning-by-doing parameters** anchored to empirical experience curve literature for Li-ion batteries (Kittner et al. 2017; Way et al. 2022)

Calibration Target: Flexibility Weight $\mu = 0.16$

From PDP8 investment allocations:

$$K_{b,ss} = \frac{2.4/10}{0.12} \approx 2.0 \text{ B USD (BESS)}$$

$$K_{g,ss} = \frac{14.9/10}{0.05} \approx 29.8 \text{ B USD (grid)}$$

With $\rho_f = 0.4$: $\mu \approx 0.16$. Reflects the *small but growing* share of private storage in Vietnam's current infrastructure.

Complementarity: $\rho_f = 0.40$

Transmission moves energy across *space*; storage moves it across *time*. When solar output is zero (night), infinite transmission capacity is useless. Hart (2019) and Hirth (2013) document strong complementarity empirically— $\rho_f = 0.4$ enforces this binding constraint.

Calibrated Parameters (I): Preferences, Production & Energy

Category	Parameter	Value	Source / Rationale
<i>Preferences & Production</i>	Discount factor β	0.990	Annualized real return $\approx 4\%$ (VGB)
	Capital share α	0.350	Labor income share = 65% (GSO 2024)
	Depreciation (prod.) δ_p	0.025	10% annually (standard)
	Labor elasticity σ_L	1.000	Frisch elasticity (NK literature)
	Energy-VA elasticity σ_E	0.600	Heutel (2012); energy–capital complementarity
<i>Energy & Reliability</i>	Reliability sensitivity ψ	2.000	Moment-matched: $u_{ss} = 97\%$ (EVN reports)
	Intermitt. persistence ρ_{ren}	0.850	Monsoon AR(1) (IEA 2023)
	Intermitt. volatility σ_{ren}	0.120	CV of monthly VN solar output (GSO 2024)
	Battery weight μ	0.160	PDP8 BESS/grid capital stock ratio
	Flexibility elasticity ρ_f	0.400	Complementarity (Hart 2019; Hirth 2013)
	Depreciation (battery) δ_b	0.030	12% annually (Li-ion lifecycle)
	Depreciation (grid) δ_g	0.0125	5% annually (transmission infrastructure)

Calibrated Parameters (II): Innovation, Shocks & External Sector

Category	Parameter	Value	Source / Rationale
<i>Learning-by-Doing</i>	Learning elasticity η_{bat}	0.100	18–22% Li-ion exp. curve (Kittner 2017; Way 2022)
	Signal transmission χ	1.0	Baseline: fully liberalized market
		0.3	Counterfactual: regulated tariff regime
<i>Battery Price Shock</i>	Persistence ρ_{bat}	0.900	AR(1) of lithium carbonate (BNEF 2023)
	Volatility σ_{bat}	0.080	Li-ion pack price survey (BNEF 2023)
<i>Fiscal Policy</i>	Grid invest. response ϕ_{grid}	1.500	Reactive lag structure (World Bank 2022)
<i>External Sector</i>	World interest rate $\bar{r}$	0.0101	$= 1/\beta - 1$ (quarterly)
	Debt premium ϕ_b	0.001	Schmitt-Grohé & Uribe (2003)
	SS net foreign assets $\bar{B}^*$	0	Balanced trade normalization

Calibration Validation: Model Fit to Vietnamese Data

Moment	Data	Model	Source
Investment rate (I/Y)	27–29%	27.2%	GSO 2024; targeted
Consumption share (C/Y)	64–66%	72.8% [†]	GSO 2024; model SS
Grid reliability (utilization)	95–98%	97.0%	EVN reports; targeted
Annualized real interest rate	$\approx 4\%$	4.04%	VGB average

[†] Note: Model abstracts from government consumption ($\approx 6\text{--}7\%$ of GDP). Adjusted private consumption share $\approx 66\%$ —consistent with data.

Reliability Parameter Validation

The steady-state utilization shortfall maps to Energy Not Served (ENS):

$$(1 - u_t) \approx \text{ENS}_t / \text{Demand}_t$$

$u_{ss} = 97\%$ implies 3% ENS—within EVN's reported 3–5% range.

Sensitivity Check

Results are robust for $\psi \in [1.5, 3.0]$ and $\mu \in [0.20, 0.35]$. The *qualitative* Agility Gap mechanism holds across the full parameter space tested.

Steady State & Solution Method

Steady-State Values

Variable	Value	Interpretation
Y_{ss}	0.868	Normalized
C_{ss}	0.631	72.8% of output
$K_{p,ss}$	8.651	Productive capital
$K_{b,ss}$	0.190	Battery capital
$K_{g,ss}$	1.140	Grid capital
u_{ss}	0.970	Grid reliability
r_{ss}^*	0.0101	World rate

Solution Method

- Model solved by **first-order perturbation** around the deterministic steady state using *Dynare 6*
- System expressed in log-deviations from steady state ($\hat{x}_t \equiv \ln X_t - \ln X_{ss}$)
- Full log-linearized system: 16 equations, 16 unknowns (see Appendix)

Blanchard-Kahn Verified

3 eigenvalues outside the unit circle match the 3 forward-looking variables (C_t , Y_t , r_t^*). Unique saddle-path equilibrium confirmed. ✓

The exponential nonlinearity of u_t is absorbed into linearization coefficient ξ . Nonlinear transition dynamics are analyzed

Baseline Results: Steady State & FEVD Overview

Forecast Error Variance Decomposition (40Q)

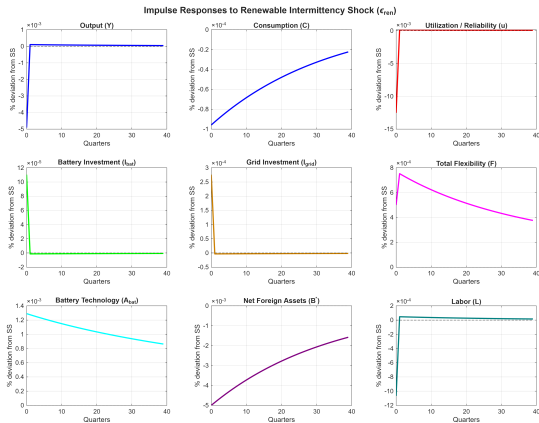
Variable	ε_{ren}	ε_{bat}	ε_I
Output (Y)	79%	16%	5%
Reliability (u)	85%	12%	3%
Consumption (C)	11%	80%	9%
Battery invest.	1%	100%	<1%
Grid invest.	14%	1%	85%

Three Structural Findings from FEVD

- 1 **Intermittency dominates** output (79%) and reliability (85%)—validates the model's core reliability channel
- 2 **Battery investment** is 100% driven by global price shocks, not domestic policy—Vietnam is a price-taker
- 3 **Consumption** is dominated by battery price shocks (80%), not intermittency—household smooths domestic shocks via borrowing but is exposed to import price volatility

Note: The B^* shares exceed 100% due to negative covariance: intermittency drains the current account (output falls) *and* battery price shocks raise import costs simultaneously.

Impulse Response: Renewable Intermittency Shock



One standard deviation shock to renewable intermittency ($\sigma_{ren} = 0.12$). All variables: % deviation from steady state over 40 quarters.

Impulse Response: Three-Stage Transmission Narrative

Stage 1 — Impact (Q0)

$u_t \downarrow -0.013\%$; $Y_t \downarrow -0.005\%$ $\Rightarrow$ Reliability penalty activates immediately via the CES output wedge.

Stage 2 — Investment Response (Q1–Q10)

$I_{bat} \uparrow +0.0001\%$ (*agile, contemporaneous*) vs. $I_{grid} \uparrow +0.0003\%$ (*sluggish, lagged*)

Agility gap is visible: private batteries respond faster than public grid. Battery import demand rises $\Rightarrow$ current account deteriorates (“twin drain”).

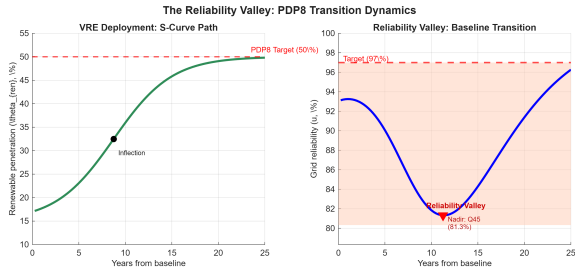
Stage 3 — Learning & Recovery (Q10+)

$A_{bat} \uparrow +0.11\%$ by Q10; endogenous LBD activated.

Gradual reliability recovery as flexibility stock accumulates. Full mean-reversion by Q30–Q35 under baseline $\chi = 1$.

Key takeaway: The intermittency shock is persistent ($\rho_{ren} = 0.85$) but the endogenous learning channel ($\chi > 0$) shortens the recovery path. Regulated markets ($\chi = 0.3$) slow recovery significantly—the counterfactual welfare cost is shown in the Appendix.

The Reliability Valley: PDP8 Transition Path (Preliminary)



*Deterministic simulation: VRE penetration 15%→50% over 100 quarters (S-curve).
Reliability falls as VRE outruns flexibility accumulation.*

Valley Statistics

Nadir at:	Quarter 45
Min. reliability:	81.3%
Below target:	15.7 pp
Implied ENS:	≈ 1,640 hrs/yr

Three Causes

- ① Front-loaded VRE (FiT boom)
- ② Lagged grid (5–7 yr permitting)
- ③ Battery supply constraints

Full welfare analysis and policy counterfactuals in progress—to be presented at final defense.